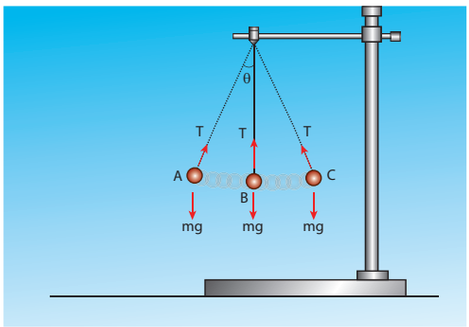
Solution

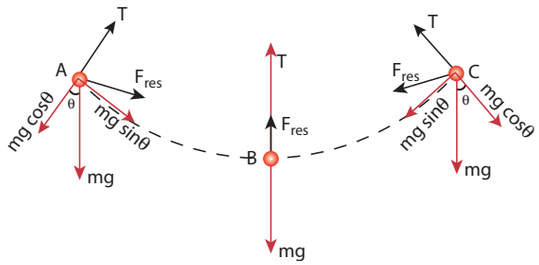
Two forces act on the bob.

(i) Gravitational force (mg) acting downwards

(ii) Tension (T) exerted by the string on the bob, whose position determines the direction of T as shown in figure.



The bob is moving in a circular arc as shown in the above figure. Hence it has centripetal acceleration. At a point A and C, the bob comes to rest momentarily and then its velocity increases when it moves towards point B. Hence, there is a tangential acceleration along the arc. The gravitational force can be resolved into two components (mg cosθ, mg sinθ) as shown below

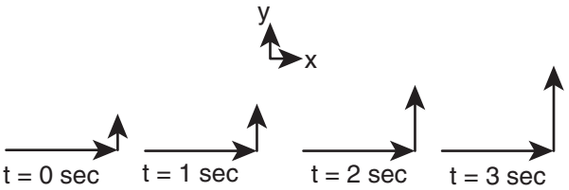


Note

Note that the bob does not move in the direction of the resultant force. At the points A and C, tension T = mg cosθ. At all other points, tension T is greater than mg cosθ, since it has non zero centripetal acceleration. At point B, the resultant force acts upward along the string. It is an example of a non uniform circular motion because the bob has both the centripetal and tangential accelerations.

EXAMPLE 3.8

The velocity of a particle moving in a plane is given by the following diagram. Find out the direction of force acting on the particle?



Solution

The velocity of the particle is $\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$. As shown in the figure, the particle is moving in the xy plane, there is no motion in the z direction. So velocity in the z direction is zero ($v_z = 0$). The velocity of the particle has x component (v_x) and y component (v_y). From figure, as time increases from t = 0 sec to t = 3 sec, the length of the vector in y direction is changing (increasing). It means y component of velocity $\left(v_y\right)$ is increasing with respect to time. According to Newton's second law, if velocity changes with respect to time then there must be acceleration. In this case, the particle has acceleration in the y direction since the y component of velocity changes. So the particle experiences force in the y direction. The length of the vector in x direction does not change. It means that the particle has constant velocity in the x direction. So no force or zero net force acts in the x direction.

EXAMPLE 3.9

Apply Newton's second law for an object at rest on Earth and analyse the result.

Solution

The object is at rest with respect to Earth (inertial coordinate system). There are two forces that act on the object.

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